Validation and AI use - frozen written record

Prepared by Peter Meixner | FIN 43900

Package built UTC 2026-10-19T20:58:39.723405+00:00. Decision date September 1, 2026. Model units: USD millions, shares millions, per-share USD. Source cutoff excludes later operating events. Full material input ledger and assumptions follow this record. The file manifest contains hashes; it does not certify human authorship.

Validation register

The pre-execution technical directional plan is docs/directional-test-plan.md. It was written by AI before final-model execution and is not the human locked prediction.

Claim / test	Result	Observed evidence
Published training-case reconciliation	PASS	All 12 published Week 3 known answers matched; per share 27.497370.
Historical articulation and normalization	PASS	Both balance sheets, cash-flow bridges and earnings reconcile. FY2025 normalized EBIT -23; H1 before restructuring 2,900.
Independent source / definition check	PASS	Official annual filing has 14,646 investing + 3,026 financing equipment payments = 17,672; issuer release independently confirms Q2 cash 12,874.
Pro-forma articulation and funding gate	PASS	21 statement columns reconcile. Low-case funding gap 5206.70 is exposed; funding gate FAIL is an intended economic finding.
Prewritten changed-input directions / monotonicity	PASS	Base 4.489388; WACC +100bp 3.116761; g +50bp 5.064008. All three cases pass signed-value directions.
Enterprise-equity and dilution bridge	PASS	1,000 cash/debt change moves equity exactly 1,000; EV unchanged. Escrow modeled in shares only.
Highest-risk failure / invalid input refusal	PASS	WACC<=g, zero shares, NaN, and +100 unsupported cash all refused.
Peer policy and multiple definitions	PASS	Both qualified peers retained; consistent annual GAAP denominators; actual annual-end shares; same September 1 unadjusted closes. INTC P/E NM.
Reverse DCF round trip	PASS	Reconstructed market price 88.970000; required EBIT margin 129.04%, economically infeasible under fixed base revenue/reinvestment.

Cold run and failure history

A genuinely fresh Python environment without pip or third-party packages and an isolated copy of source/evidence completed model.py and validate.py. results.json matched the saved result byte-for-byte. Actual command output, runtime version, timestamps and hash are in output/cold-run.log. First clean run passed; no cold-run repair was required. Earlier source retrieval failed with HTTP 403 from SEC and was resolved using Intel's official mirror. The PDF build environment is separate from the dependency-free financial engine.

The deliberate +100 cash mutation exits nonzero before valuation. Independent known answers and source definitions supplement internal checks; none proves forecast likelihood. Low-case liquidity failure remains visible as an economic finding.

Highest-risk finance claims

Terminal and bridge assumptions dominate value. Escrow is counted once; NCI and other claims remain proxies. Monotonicity uses signed equity, not the zero floor. Reverse DCF reconstructs $88.97 but its fixed-revenue 129% EBIT margin is economically infeasible.

Locked Changed-Input Record - status

New precommitted record: evidence\locked-exercise\20261019T205751Z. Precommit c6331c163b19200b728f3ad8c197d284efeacd7f; UTC 2026-10-19T20:57:51.788751+00:00; receivable days 25 to 35.0. Authorship: AI-assisted prediction and execution. Permission record: Peter Meixner reports instructor permits AI to make a realistic prediction; October 19, 2026. Not independently verified; not AI-off.. Prediction preserved before execution: AI-ASSISTED PRE-RUN PREDICTION - October 19, 2026

Prepared by Codex (GPT-6), not a human-authored AI-off prediction.

Peter Meixner reports that his professor permits AI to make a realistic prediction.

That permission is recorded as student-reported, not independently verified.

Change: base receivable days from 25 to 35 days, a 10-day increase, for the estimated pre-valuation bridge, remaining-2026 stub and FY2027-FY2031 forecasts. Units: days of annual revenue. Hold all other independent inputs unchanged. Linked receivables, liquidity, cash flow and equity value recalculate.

Prediction: operating revenue and EBIT should remain unchanged because the change affects collections, not the sales or expense assumptions. Receivables should rise, tying up additional operating cash. I expect FCFF and equity value per share to decline. The largest cash absorption should occur during the transition to the longer collection period; later incremental cash requirements should reflect changes in revenue. FY2031 FCFF should fall by tens of millions of dollars rather than billions, because only the annual change in the receivable balance is a cash-flow deduction.

I expect a modest decline in per-share value, broadly less than one dollar, rather than a collapse in operating enterprise value. The exact size remains to be tested; this estimate is not a confidence interval. Because the pre-valuation bridge is also recalculated, some of the cash effect should appear in available liquidity/bridge adjustments rather than solely in discounted future FCFF.

Expected decision effect: keep watch/defer and do not initiate at the historical $88.97 price. The change should weaken cash conversion, but the base valuation was already far below that price. If a funding-floor breach occurs, escalate to do not initiate until financing is supported. Diagnose any contrary direction, new funding failure or unusually large change rather than altering this prediction after the run.

Actual value change: -0.456618/share; final-year FCFF change -64.018213 million. Attributed diagnosis: AI-ASSISTED POST-RUN DIAGNOSIS - Codex (GPT-6), October 19, 2026.

Prepared under Peter Meixner's reported instructor permission. Not a human AI-off diagnosis.

Precommitted change: receivable days 25 to 35, +10 days. Actual equity value fell from $4.489388 to $4.032770: $-0.456618/share (-10.17%). FY2031 FCFF declined from 7525.816695 to 7461.798482 million, a change of -64.018213 million. Revenue and EBIT unchanged in every future column: True.

Enterprise value changed by -1754.438491 million; the net bridge adjustment changed by -646.004879 million. Their sum equals the change in common equity. The bridge impact occurs because the changed collection assumption also applies to the estimated period before the valuation date. It reduces liquid resources available at that date.

Mechanism: a higher receivable-days ratio requires more cash tied up in receivables. During the initial transition this absorbs cash; later changes track the growth of the receivable balance. Only the change in working capital enters FCFF. Deducting the entire receivable balance every year would overstate the effect. The actual final-year decline of about $64 million supports the pre-run prediction of tens of millions, and the roughly $0.46 per-share decline is within the broad less-than-$1 expectation. No direction reversal occurred.

Maximum funding-floor gap after change: 0.000000 million; financeable: True. Committee action remains watch/defer, with no initiation at $88.97. The unchanged action reflects an already-unmet valuation/claims gate, not an absence of financial sensitivity. These results remain conditional on the model's bridge and funding assumptions.

The old record supplies a reported human prediction before the old Lab 11 run: annual growth 5% to 6%, all five years. It predicted approximately 5% higher 2030 EBIT and $300-$400m FCFE improvement. Saved actuals: +$466.57m EBIT (+4.85%) and +$178.51m FCFE. The original text below records both the prediction and appendices. The source timestamp is a local save timestamp, not a verified partner-conversation timestamp. Immutable precommit and AI-closed execution were not established; it is retained as partial evidence, never labeled fully compliant.

Peter reported on October 19 that his professor permits an AI-generated realistic prediction. The new precommitted receivable-days experiment is labeled AI-assisted throughout, including execution and diagnosis. It does not satisfy an unmodified AI-off requirement; the instructor determines acceptance under the reported exception. Original Lab 11 gaps, absent partner exchanges, actual recordings/transcripts and access confirmations are not fabricated.

Named material AI-use register

Tool / exposed model	Material contribution	Check / disposition	Decision effect
Codex, GPT-6, October 19, 2026	Source retrieval, historical normalization and claims review	Qualify; primary filings/releases and dated Treasury/price rows; proxies explicitly labeled	Better traceability; unresolved claims still block initiation
Codex, GPT-6, October 19, 2026	FCFF statement engine, sensitivity, reverse DCF, HTML output	Accept technical implementation after known answers, identities, deliberate failures and cold run	Supplies conditional range, not forecast probabilities
Codex, GPT-6, October 19, 2026	New receivable-days prediction, precommit, execution and diagnosis	AI-assisted under Peter's reported instructor permission; committed before run; compare unchanged EBIT, cash-flow and bridge effects	Value falls from $4.4894 to $4.0328; watch/defer unchanged
Codex, GPT-6, October 19, 2026	Scenario assumptions, WACC beta/debt-cost proposals	Qualify pending student's adoption; evidence only motivates, does not verify numerical paths	Watch/defer, with downside stop condition
Earlier AI-assisted Lab 10 routine; exact model not recorded	Positive-FCFE component	Reject as a complete equity valuation; inspect max(FCFE,0), omitted terminal and missing bridge against FCFF assignment	Replace with signed operating cash-flow appraisal
Codex, GPT-6, October 19, 2026	Capex definition correction	Accept arithmetic/definition; verify 14,646 + 3,026 in separate filing classifications	Avoid understated investment requirements
Codex, GPT-6, October 19, 2026	Memo, research evolution, validation records and recording guides	Drafted for student review; numbers generated from saved model outputs	Improves consistency; no student review/recording fabricated
Prior lab AI use; model/version not available in saved records	Peer policy, model, sensitivity and presentation preparation	Disclosed in preserved lab files; do not infer unavailable version	Prior materials used with qualifications

The assistant's developer-exposed identity is GPT-6; no more precise model build ID is exposed. Tests executed by AI are labeled as such. No second-model agreement, external audit, human code review or independent student approval is asserted. Student judgment and oral defense remain the student's responsibility.

Material limitations / monitoring

The appended evidence ledger details all material limits: estimated interim balances; claim/dilution proxies; simplified compensation, NCI and residual balances; capex timing; annual peer snapshots; judgment WACC inputs; and unavailable downside financing. Reassess on earnings, financing/ownership changes or liquidity breaches. Unreconciled claims block initiation.

Preserved original Lab 11 record

Lab 11 — Locked Changed-Input Record

Company: Intel. Forecast years: 2026–2030. Cash flow measure: FCFE.

Recorded in this workspace: 2026-09-29 at 16:57 EDT, before any Lab 11 changed-input runs. This is the time the supplied notes were saved, not a claim about the time of the partner conversation. The student reports closing the laptop, working away from AI, and discussing the following with their partner. The original conversation timestamp was not supplied.

Selected driver and range

Driver: revenue growth. Base: 5% annually; lower: 4%; higher: 6%, applied to every forecast year, 2026–2030.

Reason: labelled judgment—I'm testing a recovery one percentage point slower or faster than my base case. This range is not company guidance.

Prediction supplied by the student — preserve before running

Change: annual revenue growth from 5% to 6% for 2026–2030—an increase of 1 percentage point, not 1%. All other independent assumptions stay at base.

Prediction: I expect 2030 operating profit to increase by roughly 5%, since the higher growth compounds over five years while the expense and margin ratios stay unchanged. I expect FCFE to become less negative, with a smaller dollar improvement than operating profit because taxes and working-capital needs absorb some of the gain. My rough FCFE prediction is an improvement of $300–400 million.

Mechanism: higher growth → higher revenue → higher operating profit → higher after-tax cash generation, partly offset by additional inventory and other working capital.

Partner exchange 1 — reported conversation

Partner: "You mean 6% growth in each year, correct? Why did you choose 4%–6%?"

Me: "Yes, each year. It's a narrow judgment range around my 5% recovery assumption. It tests modest variation, but it doesn't capture a return to declining sales."

Partner: "And gross margin, R&D ratios, capex and financing assumptions stay at base?"

Me: "Yes. Only revenue growth changes. Revenue, expense dollars, inventory and cash flow recalculate through the model."

Check performed on my partner's proposed analysis

Correction supplied by the student: the partner also modeled Intel. The previously reported 28%/30%/32% range was a typo. The correct FY2030 gross margins are 43% lower, 44% base and 45% higher, matching this model's final-year cases. These are final-year values, not constant margins for every forecast year. The notes below incorporate that correction.

My question, with the range corrected: "What supports your 43%–45% final-year range, and are you holding revenue growth constant so you isolate the effect of gross margin?"

Partner response, with the magnitude corrected: “The range is my judgment: one percentage point below and above the base to test weaker or stronger pricing and costs. Revenue growth and all other independent assumptions stay at base. Gross profit, taxes, inventory and cash flow recalculate through the model.”

My reported input check, corrected: the higher FY2030 margin moves from 44% to 45%, a one-percentage-point increase, and revenue growth stays unchanged.

The partner's corrected final-year margin range matches Intel's second sensitivity driver. This records the reported input check; the later numerical difference check is still pending.

Append after running — pending

- Actual 2030 operating profit and signed change from base:

- Actual 2030 FCFE and signed change from base:

- Student explanation of prediction error:

- Effect on valuation conclusion or research priority, with reason:

- Partner exchange 2: numerical difference checked, statement trace, question/correction:

- Partner exchange 3: main driver over the tested ranges, range limitation, question and response:

Remaining setup

The student must supply a second Intel operating driver, lower and higher values, units, affected years, and range rationale, and review that range with their partner before its changed-input runs. No changed-input sensitivity runs have been performed at the time this record was created.

Appended numerical result — September 29, 2026

The original prediction above is unchanged. The second driver was subsequently selected by AI at the student's request: the gross-margin path shifted by minus/plus one percentage point each year. Its assumptions are documented in README.md. The student's pre-run partner review of that second range has not been reported.

For annual revenue growth of 6% rather than 5%, FY2030 operating profit is $10,079.77 million versus $9,613.20 million at base: a signed change of +$466.57 million, approximately +4.85%. FY2030 FCFE is -$784.61 million versus -$963.12 million: a signed change of +$178.51 million. Changes use unrounded model values.

The predicted FCFE improvement was $300–400 million; the observed improvement is $121.49–221.49 million below that interval's endpoints. The student's explanation of this difference and its implications remains pending, along with the actual post-run partner exchanges.

Additional partner feedback reported by the student

My partner was initially very curious and ultimately agreed with my analysis.

This records the reported reaction. A specific post-run question and response, the numerical difference checked on the partner's model, and the student's explanation of the prediction error have not yet been supplied.

Evidence, definitions and normalization

Research assembled October 19, 2026. Decision date: September 1, 2026 regular-session close. No later operating or market facts enter the valuation. Downloaded history pages contain later dates, but the model selects only the September 1 Close column. This is a retrospective historical analysis, not a claim that this final model existed on September 1.

Primary source map

ID	Source and availability	Exact locator / use
K25	Intel FY2025 10-K; fiscal end December 27, 2025; filed January 23, 2026	Statements pp. 60-63; Note 3 segments; Note 5 shares; cash-flow financing PP&E line; Note 18 leases/commitments
Q26	Intel Q2 2026 10-Q; June 27, 2026; filed July 24, 2026	Statements pp. 3-7; Note 3 NCI; Note 4 escrow and dilution; Note 5 other liabilities; Note 11 debt
R26	Intel Q2 release, July 23, 2026, 4:01 pm EDT	Q3 GAAP guidance and independent consolidated balance sheet
AMD	AMD FY2025 release February 3, 2026 and 10-K	Annual income statement; balance sheet; actual outstanding share count
NVDA	NVIDIA FY2026 release February 25, 2026 and 10-K	Annual GAAP EPS/revenue; balance sheet p. 53 for actual shares
PRICE	Stock Analysis historical tables, retrieved October 19, 2026	September 1, 2026 row; Close, not adjusted close or current quote
UST	U.S. Treasury daily nominal yield curve	September 1, 2026 row, 10 Yr column: 4.79%

Source links:

- K25: https://www.sec.gov/Archives/edgar/data/50863/000005086326000011/intc-20251227.htm

- Q26: https://www.sec.gov/Archives/edgar/data/50863/000005086326000157/intc-20260627.htm

- R26: https://www.intc.com/news-events/press-releases/detail/1776/intel-reports-second-quarter-2026-financial-results

- AMD annual: https://ir.amd.com/news-events/press-releases/detail/1276/amd-reports-fourth-quarter-and-full-year-2025-financial-results

- AMD 10-K: https://ir.amd.com/financial-information/sec-filings/content/0000002488-26-000018/amd-20251227.htm

- NVDA annual: https://nvidianews.nvidia.com/news/nvidia-announces-financial-results-for-fourth-quarter-and-fiscal-2026

- NVDA 10-K: https://www.sec.gov/Archives/edgar/data/1045810/000104581026000021/nvda-20260125.htm

- Prices: https://stockanalysis.com/stocks/intc/history/ ; https://stockanalysis.com/stocks/amd/history/ ; https://stockanalysis.com/stocks/nvda/history/

- Treasury: https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?field_tdr_date_value_month=202609&type=daily_treasury_yield_curve

SEC was readable through the research browser but returned HTTP 403 to the evidence downloader. Intel's official filing mirror supplied the saved K25/Q26 text. The source retrieval log records URLs, times and hashes. The NVIDIA annual-end share count (24,304 million) was verified in SEC p. 53; its complete 10-K is linked rather than downloaded. Financial statement numbers below are factual data; narrative interpretations are the analyst's.

Historical fact ledger (USD millions)

Input	FY2025 K25	H1 2026 Q26	Classification / locator
Revenue	52,853	29,705	Fact; income statement
Cost of sales	34,478	17,849	Fact; income statement
R&D	13,774	6,743	Fact; income statement
MG&A	4,624	2,213	Fact; income statement
Restructuring and other charges	2,191	4,240	Fact; income statement; includes different charge categories
EBIT	-2,214	-1,340	Fact; income statement
Consolidated net income	26	-15,129	Fact; not income attributable to common shareholders
Cash / short-term investments	14,265 / 23,151	12,874 / 16,853	Fact; balance sheet
Receivables / inventory	3,839 / 11,618	4,033 / 12,492	Fact; balance sheet
Net PP&E	105,414	105,741	Fact; balance sheet
Total assets	211,429	202,439	Fact; balance sheet
Accounts payable	9,882	8,756	Fact; operating classification is a forecast simplification
Debt	46,585	50,537	Fact; short plus long-term carrying value, not market quote
Other liabilities excluding AP and debt	28,602	40,003	Derived aggregation of filing balances
Intel equity / NCI	114,281 / 12,079	87,542 / 15,601	Fact; balance sheet; NCI book is not fair value
CFO / CFI / CFF	9,697 / -14,821 / 11,587	8,102 / -669 / -8,548	Fact; signed cash-flow totals
Total ending cash incl. restricted	14,712	13,597	Fact; CF definition differs from balance-sheet cash
Restricted cash	447	723	Fact; excluded from available liquidity
Investing / financing equipment payments	14,646 / 3,026	6,192 / 1,423	Fact; both are cash equipment requirements
Depreciation	10,757	5,891	Fact; expense already embedded in operating costs

Base-year normalization and contamination catalog

Keep the original historical statements unchanged. A normalization changes the earnings interpretation; it does not rewrite reported equity or cash.

Contamination	Reconciliation / treatment	Consequence and challenge
FY2025 restructuring	Reported EBIT -2,214 + 2,191 = -23 before the restructuring line	Analytical subtotal, not recurring earnings proven by evidence; future model retains $300m annual recurring reserve

Contamination	Reconciliation / treatment	Consequence and challenge
H1 2026 restructuring/goodwill	EBIT -1,340 + 4,240 = 2,900 before that line	H1 improvement is visible; no assumption that all cash restructuring disappears
Escrow mark-to-market	H1 loss 13,619; Q2 loss 12,529 below EBIT	Not a recurring operating cost; derivative liability 15,600 converted to an equity-settlement convention, and escrow dilution included once
FY2025 consolidated income vs parent income	26 consolidated less 293 NCI = -267 parent	Positive consolidated income cannot rescue Intel's reported P/E; EPS -0.06 remains NM
Altera scope change	FY2025 includes pre-sale activity; deconsolidated September 2025	Do not call reported growth organic; reset near-term revenue using H1 2026 and Q3 guidance
Segment transfers	Internal foundry revenue is eliminated on consolidation	Use consolidated sales; never add reported foundry revenue to processor revenue without eliminations
Capex field definition	FY2025 cash equipment = 14,646 + 3,026 = 17,672	A provider investing-only field misses 3,026; future $18bn cash/additions proxy assumes stable vendor financing
Asset additions vs cash payments	H1 also reports noncash PP&E acquisitions	Future cash capex = asset additions is a steady-payables assumption, not an identity of historical accounting
Restricted cash	FY2025 14,712 - 447 = 14,265; H1 13,597 - 723 = 12,874	Restricted balances are not added as available bridge cash
Ireland SCIP buyout	Already embedded in June actual assets, financing and equity	Do not deduct the purchase price again; exclude Ireland NCI after the buyout
SBC / GAAP margins	Forecast expenses retain compensation; no SBC cash add-back	Cash-equivalent replacement-pay assumption avoids free compensation and unmodeled perpetual dilution
Tax valuation allowances	Historical effective tax rate is distorted	Forecast 21% on positive income; zero current credit for losses; no separate NOL value
Dates and shares	July 17 basic shares 5,044; annual EPS shares 4,530	Use a separate 5,257m diluted-equivalent valuation proxy; annual EPS denominator is not current capitalization

Forecast, timing and statement conventions

Q3 guidance was $15.8-$16.8bn revenue and 41% GAAP gross margin, available July 23. Base H2 revenue of $32.6bn repeats the midpoint for Q4: Q4 is an analyst judgment, not guidance. Full FY2026 base sales are H1 actual $29.705bn plus H2 estimate $32.6bn. R&D and SG&A remain separate; manufacturing margin and fab capex carry the central turnaround risk. The model does not grant external foundry sales beyond consolidated scenario growth.

The starting balance sheet is June 27. A 66-day estimated operating bridge reaches September 1; a 116-day remaining stub reaches December 26. Only the remaining stub is discounted, followed by five complete annual forecasts FY2027-FY2031. Pre-valuation estimates are exposed in results.json and never presented as actual reported balances. No post-June capital transaction is assumed unless disclosed in the included evidence; unverified intervening changes are an initiation-blocking limitation. Subsequent years use annual intervals, a disclosed fiscal-calendar approximation.

FCFF = EBIT - modeled unlevered taxes + depreciation/amortization - cash capex - change in operating NWC. Debt repayments, investment sales and NCI distributions are excluded from FCFF. Negative cash flows are retained. In the cash-flow statement, interest and actual modeled tax enter CFO; a cash-equivalent compensation convention is used. PP&E and intangibles roll forward independently. The balance sheet separately carries common and minority equity. NCI income is 10% of positive consolidated profit and distributed in full; this is a placeholder allocation, not subsidiary-specific economics.

Other assets/liabilities are held flat except escrow normalization; receivables, inventories and payables move with activity. Operating leases stay in operating expenses. The $5bn other-claims bridge reserve comprises $3bn deferred compensation and a $2bn judgment allowance for unresolved financing/lease/vendor claims. These reserves do not claim a verified fair-value total. No lease-principal cash flow is double deducted from FCFF. Exact vendor/lease settlement schedules remain a material limitation.

Enterprise-to-equity definitions

Add estimated September 1 cash above a $5bn operating minimum, remaining short-term investments and $8.410bn equity investments at carrying-value proxy. Subtract modeled outstanding debt, $15.601bn NCI at book-value proxy, and $5bn other claims. Consolidated FCFF includes Mobileye/Arizona/IMS operating economics, so do not add full subsidiary equity values again. No cash beyond the opening bridge is separately added later. Cash spent on forecast losses is captured in signed FCFF; adding opening liquidity is not double counting.

Diluted-equivalent shares = 5,044 basic + 143 escrow + 70 awards = 5,257 million. The 143 is the filing's total unreleased escrow, not the rounded 71 + 71 subtotals. The 70 anti-dilutive share disclosure is used conservatively as an award proxy, with possible overlap/vesting uncertainty; it is not a treasury-stock-method fully diluted certification. No escrow derivative deduction accompanies this share-inclusive approach. Contingent 241m government warrants are excluded absent loss of foundry control; a separate exercise what-if adds shares and $4.82bn strike proceeds, holding operating economics constant only to isolate dilution.

WACC and terminal convention

Ke = 4.79% Treasury + 1.50 judgment beta x 5% course ERP = 12.29%. Debt cost 6.5% is a marginal borrowing judgment; it is not historical interest/debt. Market equity uses $88.97 x 5,044m basic shares; June debt is a carrying-value proxy for market debt. After-tax Kd uses 21%, giving approximately 11.57% WACC. Low/high cases shift WACC +2/-1 percentage points. Beta, debt fair value and realizable tax shields remain assumptions; sensitivity does not verify them.

End-period discounting; constant-growth terminal value uses FY2031 FCFF x (1+g)/(WACC-g). Base g=2.5%; low 1.5%; high 3%. No terminal step-up is used. This retains final-year net reinvestment rather than assuming capex equals depreciation. The output reports the implied steady-state reinvestment return as a plausibility diagnostic. Final-year consolidated amortization reaches zero as the current intangible balance amortizes; no new acquisitions are assumed. WACC <= g and nonpositive terminal FCFF are refused.

Comparison definitions and exclusions

P/E uses annual GAAP diluted EPS; Intel's negative EPS makes the target method unusable. EV/sales uses September 1 unadjusted prices and annual-end actual shares/balance-sheet adjustments for all companies; it is explicitly a lagged annual-balance reference. NVIDIA nonmarketable investments and Intel equity investments are subtracted as nonoperating holdings. AMD has no separately material investment adjustment in this simplified reference. Minority and contingent claims follow the disclosed rows in inputs.json. Operating lease costs remain operating; there is no lease-capitalized EBITDA comparison.

Large peer multiples reflect profitable fabless/accelerator growth businesses and differ fundamentally from an integrated manufacturer's cash requirements. A revenue multiple ignores the very capex and margin risks driving Intel's DCF. The diagnostic implied-price interval is shown but rejected as a committee valuation range. No supported transaction multiple is used. No merger premium, liquidation value, tax-asset option, government guarantee, future subsidy receipts or speculative foundry spin-off value is asserted.

Assumption-challenge record

All numerical forecast choices are AI-assisted proposals for student review, not management guidance unless explicitly identified. Values are USD millions unless percentages or days are shown. Paths include the 2026 stub, then 2027-2031. Cases are causal packages rather than independent extremes multiplied together.

Load-bearing driver	Value / basis	Challenge	Evidence that would change it
H2 sales	Low/base/high 30,600 / 32,600 / 34,600; base repeats Q3 guidance midpoint	Q4 may be seasonal or capacity-constrained; high exceeds repeated guidance	Q3 reported revenue outside 15,800-16,800; updated backlog/output and Q4 guidance
Base annual growth	8%, 6%, 5%, 4%, 3%; server recovery fades toward maturity	Gains may be share loss reversal, price or scope rather than sustainable volume	Segment demand, units/pricing, shipment capacity and two consecutive revenue misses
Low/high growth	Low 0,2,3,3,3%; high 12,10,8,6,5%	No probability is attached; not a statistical confidence band	Customer commitments, market-share evidence and utilization
Gross margin	Base 41,42,44,45,46,46%; low 38 to 42%; high 43 to 52%	Factory yields/cost absorption may prevent recovery; endpoints exceed recent history	Two quarters below 38% trigger downside review; 41%+ and improving manufacturing losses support continued watch
R&D ratio	Base 22.5% to 20%; H1 ratio 22.7%	Cost restraint can damage process/product roadmap	Roadmap delays, R&D dollars vs hiring/program commitments; raise ratio if recovery requires more spend
MG&A ratio	Base 7.5%; low 8%; high 7%	Current savings may not persist	Reconciled GAAP expense run rate and restructuring cash needs
Recurring restructuring	300 annually, retained in all cases	Repeated restructuring may exceed this	New charge plans, cash settlement footnotes; increase reserve
Cash capex	Base 18,000 annually; low 18-19bn; high 20-22bn	New nodes and customer commitments may require more cash; cash/additions timing differs	Equipment commitments, financing PP&E payments, new lease obligations; +4bn what-if
Depreciation	11,782 / 105,741 of opening PP&E per year	Asset mix, useful lives and idle assets may change	PP&E life revisions, impairments, assets entering service
Amortization	938 annual until 2,545 existing intangible balance exhausted	New acquisitions would replenish balance	Acquisitions and acquired-intangible schedules
Inventory days	Base 125, low 140, high 115	Higher inventory can hide demand weakness; end-balance rather than average	Inventory composition, turns, aging, write-downs; revise if >140 days
Receivable/payable days	Base 25 / 85; stress cases in inputs	AP may include equipment payables; collection and payment terms can move	Receivable aging, factoring, vendor-financing notes
Cash minimum / funding	5,000; sell existing ST investments at par; no new debt or equity	A major fab may need more liquidity; not all holdings may be liquid	Restrictions, committed liquidity, approved financing terms; minimum-liquidity breach stops initiation
Debt cost / repayment	6.5%; 2,000 annual repayment scenario	Not a bond-by-bond maturity schedule; could force refinancing	Verified maturity ladder and traded yields; higher costs or near-term concentration
Taxes	21% on positive pretax/unlevered EBIT, no loss refunds	NOLs may defer taxes; global effective rate differs	Cash-tax jurisdiction detail and realization evidence
NCI income / distribution	10% positive consolidated NI, distributed; no minority loss allocation	Not subsidiary-level; 49% Arizona/23% Mobileye differ materially	Subsidiary cash-flow and contractual distribution forecasts

Load-bearing driver	Value / basis	Challenge	Evidence that would change it
NCI bridge	15,601 book proxy; +50% stress	Fair value can differ greatly; book value is not evidence of market value	Listed subsidiary equity prices and Arizona partner economics
Equity investments	8,410 book proxy; zero-value stress	Monetization, tax leakage and restrictions	Holdings detail, observable prices, tax basis and sale terms
Other claims	5,000 reserve: 3,000 deferred comp plus 2,000 unresolved financing/lease allowance	Incomplete fair-value schedule; reserve cannot certify completeness	Contract-level lease/vendor/employee claims reconciliation before initiation
Share basis	5,257 diluted-equivalent proxy	Anti-dilution disclosure is not a current option valuation; possible overlap	Equity roll-forward, award schedules, government contingencies; 241m warrant stress
WACC beta / ERP	1.50 judgment / 5% course assumption; Rf 4.79% dated	Beta is not estimated from a regression; cap structure may shift	Reproducible unlevered peer beta and current yield/capital structure evidence
Terminal growth	Base 2.5%, low 1.5%, high 3%	Final-year reinvestment/margins must support perpetual growth	Stable capacity economics, ROIC and cash-conversion evidence; TV/EV reviewed explicitly
Held-flat residual balances	Other assets/liabilities flat; no extra subsidy or acquisitions	Taxes, deposits and deferred obligations can move materially	Detailed roll-forwards; any material change requires rebuilding operating NWC/bridge
Cash-equivalent compensation	SBC stays in expense; no add-back or ongoing share issuance	Cash replacement differs from actual employee equity plan	Full SBC/dilution model; do not add back SBC without a corresponding equity cost

Reported human fresh-eyes evidence, preserved with attribution limits

The existing Intel-Lab10/README.md records the student's report of a partner discussion: the partner challenged a 44% terminal margin against a three-year historical maximum of 40.04%. The student's recorded response described utilization/cost recovery as a scenario and promised revision if manufacturing costs remained elevated. Credit: **Sam Stewart**, identified by Peter Meixner on October 19, 2026. This is reported prior feedback, not a newly witnessed conversation. The final base endpoint is 46%, so the challenge remains unresolved and is stronger, not closed by the newer model. Current Q3 guidance supports only 41%, not the long-run endpoint.

The same file records the student's challenge to the partner about growing sales with flat capex. The final model retains $18bn base cash investment, increases high-case investment, and explicitly tests a $4bn annual overrun. This is a model response to the recorded issue, not proof of sufficient capacity.

The original Lab 11 record reports a 5%-to-6% growth prediction and later actual FCFE difference. It also reports that the second margin range was AI-selected and that specific post-run partner exchanges were missing. Those gaps remain disclosed; no conversations or independent student conclusions have been invented.

Fresh technical review in this session

Codex self-review identified stale-date cash-flow timing, signed-FCFE truncation, annual EPS shares used as current shares, missing financing equipment payments, and possible escrow double counting as principal failure modes. Changes: explicit remaining-year stub, signed FCFF, separate share proxy, source-definition check, and an equity-settled escrow convention. These are AI-assisted checks, not a second human review.